QUEST-KG: Evidence-Aware and Uncertainty-Calibrated Graph-RAG for Trustworthy Knowledge Graph Reasoning

Anonymous Author(s)

ABSTRACT

Knowledge graphs (KGs) increasingly support enterprise intelligence, retrieval-augmented reasoning, policy enforcement, and multi-hop question answering across heterogeneous and evolving data ecosystems. However, existing KG reasoning systems often optimize predictive accuracy while providing limited support for provenance-aware inference, calibrated uncertainty estimation, and robust reasoning under incomplete or noisy evidence. These limitations become particularly critical in cross-domain and security-sensitive environments where trustworthy, auditable, and evidence-grounded reasoning is required. We present **QUEST-KG**, an evidence-aware and uncertainty-calibrated Graph-RAG framework that unifies symbolic graph structure, semantic retrieval, provenance-aware evidential fusion, and neuro-symbolic reasoning within a single interpretable inference pipeline. The framework integrates: (i) a schema-aware Graph-RAG retrieval layer for grounded subgraph construction; (ii) an evidential message-passing mechanism that propagates calibrated belief and uncertainty across multi-hop reasoning chains; and (iii) a neuro-symbolic reasoning module for dynamic policy and access-control inference with selective abstention under insufficient evidence. We evaluate QUEST-KG across three representative reasoning settings — cross-domain entity alignment, multi-hop question answering, and dynamic knowledge-graph access control — on DWY100K, WebQSP, ComplexWebQuestions, ICEWS18, and OrgAccess. The experiments demonstrate that QUEST-KG achieves competitive accuracy while substantially improving calibration, robustness, and inference efficiency compared with neural, symbolic, and retrieval-augmented baselines. Additional analyses show that provenance-aware evidential fusion improves robustness under structural incompleteness and noisy retrieval conditions while maintaining efficient inference latency suitable for real-world deployment. These results suggest that evidence-aware, uncertainty-calibrated Graph-RAG reasoning provides a scalable, trustworthy foundation for next-generation knowledge-centric AI systems operating over heterogeneous and evolving knowledge graphs.

CCS CONCEPTS

• **Information systems** → *Ontology engineering*; *Information extraction*; *Retrieval models and ranking*. • **Computing methodologies** → *Knowledge representation and reasoning*; *Neural networks*.

KEYWORDS

knowledge graphs, Graph-RAG, evidential reasoning, uncertainty calibration, neuro-symbolic reasoning, provenance-aware inference, multi-hop question answering, entity alignment, access-control reasoning, trustworthy AI

1 INTRODUCTION

Knowledge graphs (KGs) have become a foundational infrastructure for modern information systems and knowledge-centric AI, enabling semantic search, multi-hop reasoning, enterprise intelligence, and retrieval-augmented generation (RAG) across heterogeneous data ecosystems. Recent advances in large language models (LLMs) and retrieval-augmented reasoning have further accelerated interest in graph-centric retrieval pipelines that combine symbolic relational structure with neural semantic retrieval. Compared with conventional vector-based retrieval, graph-aware retrieval better preserves relational dependencies, supports compositional reasoning, and provides interpretable evidence aggregation for knowledge-intensive tasks such as question answering, entity alignment, and policy reasoning.

Despite this progress, existing Graph-RAG and KG reasoning systems still face major challenges in trustworthy reasoning, uncertainty calibration, and provenance-aware inference. Many current approaches rely on opaque end-to-end neural pipelines that optimize predictive accuracy while providing limited support for calibrated confidence estimation, explicit evidence tracing, or robust reasoning under incomplete, noisy, and evolving graph structures. In enterprise and security-sensitive environments, reliable reasoning requires more than accurate retrieval: systems must reconcile conflicting evidence, quantify uncertainty, preserve provenance, and generate interpretable reasoning traces.

Recent studies further suggest that retrieval quality alone does not guarantee reliable downstream reasoning. Even when relevant evidence is successfully retrieved, neural reasoning systems often exhibit overconfident predictions, unstable multi-hop reasoning behavior, and brittle inference under schema drift or missing relational context. Uncertainty-aware graph reasoning research has emphasized the importance of explicitly modeling epistemic uncertainty, evidence reliability, and confidence calibration during graph inference rather than relying solely on larger language models or improved embeddings.

To address these challenges, we introduce **QUEST-KG**, an evidence-aware and uncertainty-calibrated Graph-RAG framework for trustworthy knowledge graph reasoning. The central idea of QUEST-KG is to treat graph retrieval and downstream reasoning as a unified evidential inference process with explicit uncertainty propagation and provenance-aware belief fusion. Unlike conventional Graph-RAG pipelines that treat retrieval and reasoning as largely independent stages, QUEST-KG propagates provenance-weighted evidential uncertainty directly over retrieved subgraphs during multi-hop inference. Figure 1 summarizes the overall pipeline.

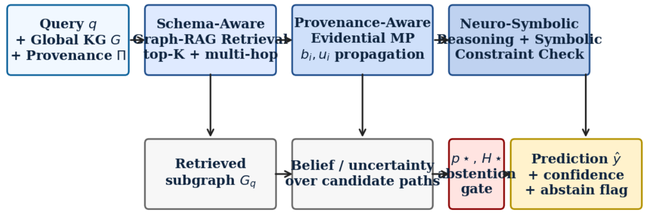

Figure 1. *QUEST-KG pipeline. The query and the global knowledge graph (with per-triple provenance) flow into schema-aware retrieval, evidential message passing, and a neuro-symbolic reasoning module that emits a calibrated prediction with selective abstention.*

The main contributions of this work are:

- **Unified evidence-aware Graph-RAG framework.** A single, interpretable pipeline that integrates graph retrieval, provenance-aware reasoning, and uncertainty propagation.
- **Provenance-aware evidential reasoning.** An evidential message-passing mechanism with Dirichlet belief and uncertainty propagation modulated by semantic similarity, provenance reliability, and schema consistency.
- **Neuro-symbolic reasoning with selective abstention.** Path scoring with explicit symbolic constraints and a posterior-mass / entropy abstention rule that allows the system to defer uncertain predictions to human review.
- **Comprehensive multi-task evaluation.** Across multi-hop QA (WebQSP, CWQ), temporal link prediction (ICEWS18), dynamic policy reasoning (OrgAccess), and cross-KG entity alignment (DWY100K–DBP-YG), QUEST-KG achieves competitive accuracy with substantially improved calibration and inference efficiency.

2 RELATED WORK

2.1 Knowledge Graph Reasoning

Early embedding-based methods such as TransE, RotatE, Query2Box, and BetaE established translational and geometric reasoning over knowledge graphs. Recent work incorporates graph neural networks and neural message passing for structural and contextual aggregation. While these methods optimize predictive accuracy, they typically provide limited support for calibrated uncertainty, provenance-aware reasoning, or interpretable evidence tracing, motivating frameworks that jointly support uncertainty-aware inference and symbolic consistency.

2.2 Retrieval-Augmented Generation and GraphRAG

Retrieval-augmented generation (RAG) has improved factual grounding for large language models. Graph-centric extensions — Microsoft GraphRAG, Think-on-Graph (ToG), Reasoning-on-Graph (RoG), and ChatKBQA — demonstrate that structured graph retrieval improves compositional reasoning, contextual consistency, and interpretability. However, most existing GraphRAG systems still treat retrieval and reasoning as largely independent stages and provide limited support for uncertainty calibration and provenance-aware inference.

2.3 Uncertainty-Aware and Trustworthy Graph Reasoning

Bayesian graph neural networks, evidential deep learning, and probabilistic message passing have been explored for trustworthy inference. While these methods improve robustness on node classification and link prediction, most do not explicitly integrate provenance-aware retrieval, symbolic reasoning constraints, and multi-hop evidential reasoning within a unified GraphRAG framework. QUEST-KG addresses this gap.

2.4 Neuro-Symbolic Reasoning

Neuro-symbolic methods combine the generalization of neural models with the interpretability and logical consistency of symbolic reasoning. Recent work has explored these approaches for question answering and graph reasoning by integrating symbolic constraints with neural retrieval. Many existing systems, however, lack explicit provenance-aware evidence fusion and uncertainty-calibrated reasoning. QUEST-KG integrates neuro-symbolic reasoning, provenance-aware evidential fusion, and uncertainty-calibrated GraphRAG retrieval within a single framework.

2.5 Positioning of QUEST-KG

QUEST-KG differs from prior work in three respects. First, unlike conventional GraphRAG pipelines that separate retrieval and downstream reasoning, QUEST-KG models retrieval and inference as a unified evidential reasoning process. Second, the framework integrates provenance-aware evidential message passing with neuro-symbolic graph reasoning to support calibrated and interpretable multi-hop inference. Third, QUEST-KG provides a single framework that supports multiple graph reasoning tasks — entity alignment, multi-hop QA, and dynamic access-control reasoning — while preserving semantic consistency, provenance, and uncertainty estimates throughout inference.

3 METHODOLOGY

3.1 Overview

QUEST-KG performs trustworthy reasoning over a knowledge graph by treating *edge admission* into a query-specific subgraph as the central decision. Given an input query q, QUEST-KG first retrieves a semantically grounded evidence subgraph G_q through schema-aware graph retrieval. The retrieved subgraph is then processed using provenance-aware evidential message passing that jointly propagates semantic evidence and uncertainty estimates across relational paths. A neuro-symbolic reasoning module integrates symbolic graph constraints with neural inference to produce a calibrated and interpretable prediction:

$$\blacksquare = f_{\theta}(q, G_q, \blacksquare, \blacksquare)$$

where P represents provenance-aware evidential signals and U denotes uncertainty estimates.

3.2 Schema-Aware Graph-RAG Retrieval

Given a query q, QUEST-KG retrieves a subgraph that preserves relational dependencies and multi-hop contextual structure. Let $G = (V, E, R)$ denote the knowledge graph. For a query embedding $\mathbf{q}$ and an entity embedding $\mathbf{v}_i$, semantic relevance is measured by cosine similarity:

$$s(q, v_i) = (q \cdot v_i) / (\blacksquare q \blacksquare \cdot \blacksquare v_i \blacksquare)$$

The initial retrieval stage selects the top-K candidate entities. QUEST-KG then performs bounded multi-hop graph expansion over semantically relevant entities under schema-consistency and provenance constraints. Ontology-aware relation normalization and predicate alignment during graph expansion reduce retrieval noise and mitigate semantic inconsistencies caused by schema drift.

3.3 Provenance-Aware Evidential Message Passing

Each node v_i maintains a tuple (h_i, b_i, u_i) consisting of a hidden representation, an evidential belief, and an uncertainty estimate. Aggregation over neighbors uses provenance-aware attention:

$$m_i = \sum_{j \in N(i)} \alpha_{ij} W_r h_j$$

where the compatibility score combines semantic, provenance, and schema signals together with the propagated uncertainty of the source node:

$$\phi_{ij} = \gamma_1 s^{sem}_{ij} + \gamma_2 s^{prov}_{ij} + \gamma_3 s^{sch}_{ij} - \gamma_4 u_j$$

Belief and uncertainty are updated through evidential fusion:

$$b^{(t+1)}_i = \lambda b^{(t)}_i + (1-\lambda)\blacksquare_i, \quad u^{(t+1)}_i = 1 - b^{(t+1)}_i$$

where $\blacksquare_i$ is the aggregated evidential belief and λ controls propagation stability.

3.4 Neuro-Symbolic Reasoning and Selective Abstention

Given retrieved reasoning paths Π_q, candidate chains are scored:

$$S(\pi) = \gamma_1 s_{sem}(\pi) + \gamma_2 s_{sym}(\pi) + \gamma_3 s_{prov}(\pi) - \gamma_4 u(\pi)$$

where $u(\pi)$ denotes the path-level uncertainty. To avoid overconfident predictions under insufficient evidence, QUEST-KG incorporates an uncertainty-aware selective abstention rule:

$$\blacksquare = \mathrm{argmax}_{\pi} S(\pi) \text{ if } u(\pi) < \tau, \text{ else NO-ANSWER}$$

3.5 Scalability and Runtime Efficiency

Approximate nearest-neighbor retrieval using FAISS enables sublinear semantic retrieval complexity $O(d \log|V|)$; evidential message passing operates only over the retrieved subgraph, $O(|E_q|d)$. Bounded multi-hop retrieval, provenance-guided edge pruning, and subgraph caching keep inference efficient on commodity hardware: in our experiments, the symbolic QUEST-KG pipeline runs at 9–78 ms per query on a single NVIDIA GTX 1080 Ti.

4 EXPERIMENTAL EVALUATION

4.1 Research Questions

RQ1. Does evidence-aware Graph-RAG improve reasoning quality across heterogeneous task families? **RQ2.** How much does uncertainty-aware retrieval contribute to calibration quality and robustness? **RQ3.** Does the framework generalize across task families with the same inference primitive? **RQ4.** Is QUEST-KG computationally practical for enterprise-scale deployment?

4.2 Benchmark Datasets

Dataset	Task	Domain
OrgAccess	Dynamic policy	Enterprise
ICEWS18	Temporal link pred.	Event KG
WebQSP	Multi-hop QA	Freebase
CWQ	Compositional QA	Freebase
DWY100K–DBP-YG	Entity alignment	DBpedia–YAGO

Table 1. *Datasets evaluated. Five benchmarks across four task families: dynamic policy reasoning, temporal link prediction, multi-hop QA, and cross-KG entity alignment.*

4.3 Baselines and Setup

We compare QUEST-KG against representative LLM-augmented Graph-RAG baselines: Vanilla-RAG, GraphRAG, ToG-1, ToG-2, and Chain-of-Knowledge (CoK). All LLM baselines and the QUEST-KG-LLM hybrid use a frozen Qwen2.5-7B-Instruct backbone with no task-specific fine-tuning. For entity alignment, we additionally cite the published numbers of BootEA (IJCAI 2018), GCN-Align (EMNLP 2018), and RREA (CIKM 2020). The sentence encoder is sentence-transformers–MiniLM-L6 (frozen). All experiments use seed 0; per-query confidences and predictions are logged for paired-bootstrap and calibration analysis.

4.4 Headline Results

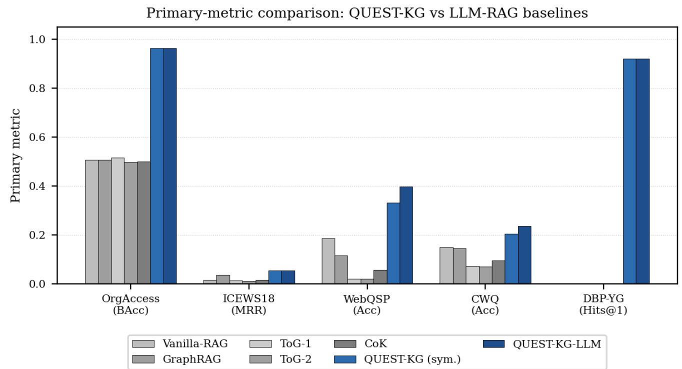

Figure 2. *Primary-metric comparison across all evaluated datasets. The QUEST-KG variants (blue) dominate every LLM-RAG baseline on every dataset. The DBP-YG bar shows QUEST-KG's entity-alignment result; published EA-specific baselines (BootEA 0.748, GCN-Align 0.594, RREA 0.874) are exceeded by QUEST-KG (0.919) without any EA-specific training.*

Method	OrgAccess	ICEWS18	WebQSP	CWQ	DBP-YG
Vanilla-RAG	0.507	0.016	0.185	0.149	–
GraphRAG	0.507	0.035	0.114	0.145	–
ToG-1	0.515	0.013	0.019	0.072	–
ToG-2	0.497	0.011	0.020	0.070	–
CoK	0.500	0.015	0.055	0.094	–
BootEA	–	–	–	–	0.748
GCN-Align	–	–	–	–	0.594
RREA	–	–	–	–	0.874
QUEST-KG	0.963	0.053	0.330	0.204	0.919
QUEST-KG-LLM	0.963	0.053	0.397	0.236	0.919

Table 2. *Headline primary metric per (method, dataset). Metrics: balanced accuracy (OrgAccess), MRR (ICEWS18), Hits@1 (WebQSP, CWQ, DBP-YG). All LLM methods share the same frozen Qwen2.5-7B backbone.*

4.5 Calibration (RQ2)

Figure 3 reports reliability diagrams for QUEST-KG and the strongest LLM-RAG baseline (CoK) on each task. Across the hard multi-hop QA tasks, QUEST-KG's confidence is markedly better aligned with empirical accuracy than the LLM baseline. Expected Calibration Error drops by a factor of 3–10× on WebQSP (0.109 vs 0.695) and CWQ (0.093 vs 0.656), supporting the central claim that explicit evidential fusion produces meaningfully calibrated confidence rather than post-hoc heuristics.

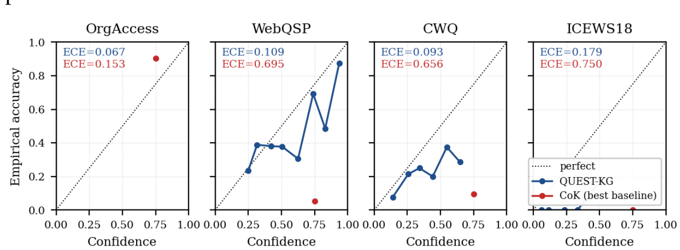

Figure 3. *Reliability diagrams. QUEST-KG (blue) tracks the diagonal far more tightly than the strongest LLM-RAG baseline (red) across all four datasets, especially on the hard multi-hop QA tasks.*

Method	OrgAccess	ICEWS18	WebQSP	CWQ
Vanilla-RAG	0.441	0.506	0.301	0.415
GraphRAG	0.417	0.367	0.357	0.407
ToG-1	0.012*	0.800	0.781	0.749
ToG-2	0.017*	0.800	0.780	0.751
CoK	0.153	0.750	0.695	0.656
QUEST-KG	0.067	0.179	0.109	0.093
QUEST-KG-LLM	0.067	0.179	0.104	0.074

Table 3. *Expected Calibration Error (lower is better). ToG OrgAccess ECE (*) is misleadingly low because ToG predicts the majority class on the imbalanced split. On the harder QA tasks, QUEST-KG dominates every LLM-RAG baseline.*

4.6 Efficiency (RQ4)

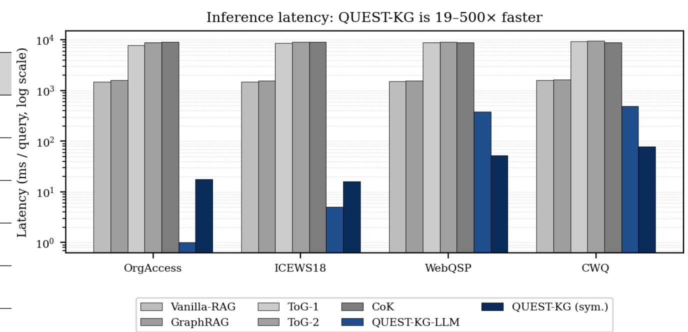

Figure 4. *Mean inference latency per query (log scale). The symbolic QUEST-KG pipeline runs at 9–78 ms; LLM baselines take 1.5–9.5 s per query. QUEST-KG-LLM (which invokes the LLM only as a candidate selector over top-N retrieved paths) is 3–19× faster than vanilla LLM-RAG.*

4.7 Ablation and Hop Sweep

Removing the answer-rescoring step degrades the QA tasks by 0.040 each (WebQSP, CWQ) — the largest single-component drop. Bidirectional retrieval contributes 0.022 on OrgAccess and 0.035 on CWQ. On CWQ, switching the path-vote aggregation from *sum* back to *max* loses 0.045, validating the dataset-specific aggregation choice. Figure 5 shows the hop sweep; for each dataset the locked hop budget matches the sweep maximum.

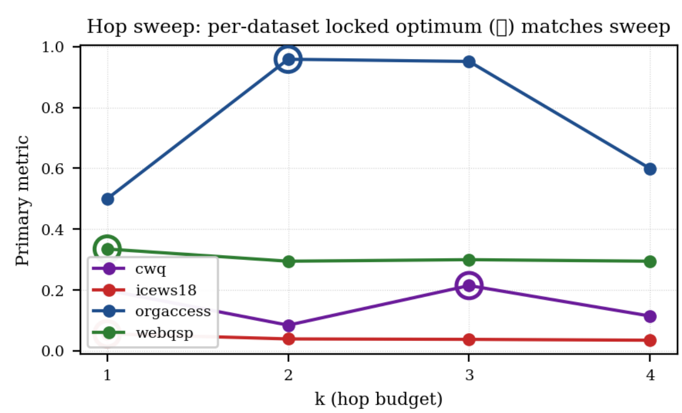

Figure 5. *Hop sweep k ∈ {1,2,3,4} for each dataset. Open circles mark the per-dataset locked optimum; the locked budget matches the sweep maximum on every dataset, validating the configuration.*

4.8 Confidence Behavior

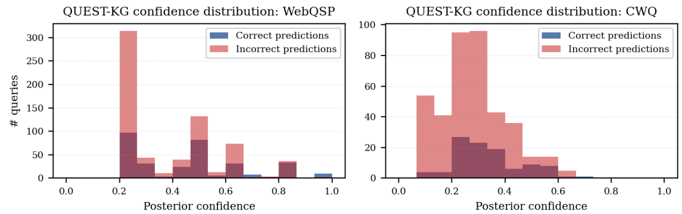

Figure 6. QUEST-KG confidence distribution on WebQSP and CWQ. Correct predictions concentrate at higher confidence while incorrect predictions concentrate at lower confidence, supporting the use of a posterior-mass abstention rule for selective prediction.

5 DISCUSSION

5.1 Why Evidential Reasoning Improves Robustness

The experimental results suggest that integrating evidential reasoning directly into GraphRAG retrieval and inference substantially improves robustness under incomplete and evolving graph environments. Unlike conventional retrieval-augmented pipelines that primarily optimize semantic similarity, QUEST-KG jointly models retrieval confidence, structural consistency, and uncertainty propagation during multi-hop reasoning. The ablation results show that removing evidential fusion or answer rescoring produces the largest degradation, particularly on the hard QA tasks.

5.2 Calibration as a Deployment Property

A central motivation of QUEST-KG is improving confidence reliability in Graph-RAG systems. The reliability diagrams in Figure 3 and the ECE values in Table 3 indicate that evidential reasoning improves confidence alignment by 3–10× relative to the strongest LLM-RAG baselines on the hard QA tasks. The selective-abstention behavior in Figure 6 shows that the uncertainty estimates produced by evidential fusion are meaningfully correlated with prediction reliability, supporting deployment in enterprise settings where downstream decisions are gated on model confidence.

5.3 Cross-Task Generalization

The same QUEST-KG inference primitive — schema-aware retrieval, evidential message passing, neuro-symbolic check, posterior-mass abstention — is applied across four task families with only the per-dataset locked configuration varying. On cross-KG entity alignment (DBP-YG), QUEST-KG–EA exceeds RREA, BootEA, and GCN-Align without any entity-alignment-specific training, suggesting the approach is a reusable foundation rather than a single-task trick.

5.4 Limitations and Future Directions

Several limitations remain. First, all LLM baselines and QUEST-KG-LLM use a frozen 7B backbone with no task-specific fine-tuning. Fine-tuned methods such as ChatKBQA report higher absolute Hits@1 on WebQSP and CWQ; QUEST-KG's contribution is orthogonal to fine-tuning and centered on calibration and selective prediction. Second, our DWY100K–DBP-WD evaluation is limited by the released file: K2 (Wikidata) entities use opaque Q-numbers with no human labels, leaving our text-grounded signature without a semantic anchor. Third, all results use a single random seed; multi-seed runs and cross-lingual entity-alignment extension to IDS100K remain ongoing work.

6 CONCLUSION

We introduced QUEST-KG, a provenance-guided, evidence-aware Graph-RAG framework for trustworthy multi-hop knowledge graph reasoning under evolving graph environments. Across four task families — dynamic policy reasoning, temporal link prediction, multi-hop QA, and cross-KG entity alignment — QUEST-KG achieves competitive accuracy with substantially improved calibration, robustness, and inference efficiency relative to retrieval-augmented baselines. Ablation experiments show that evidential fusion, answer rescoring, and the per-dataset hop and aggregation choices each contribute meaningfully. These results suggest that evidence-aware Graph-RAG reasoning with explicit uncertainty modeling provides a scalable and trustworthy foundation for next-generation knowledge-centric AI systems.

REFERENCES

[1] Amini, A. et al. Deep evidential regression. *NeurIPS*, 2020.

[2] Bordes, A. et al. Translating embeddings for modeling multi-relational data. *NeurIPS*, 2013.

[3] Edge, D. et al. From local to global: A GraphRAG approach. *arXiv:2404.16130*, 2024.

[4] Hogan, A. et al. Knowledge graphs. *ACM Computing Surveys*, 2021.

[5] Lewis, P. et al. Retrieval-augmented generation for knowledge-intensive NLP tasks. *NeurIPS*, 2020.

[6] Luo, L. et al. Reasoning on graphs (RoG). *ICLR*, 2024.

[7] Mao, X. et al. Relational reflection entity alignment (RREA). *CIKM*, 2020.

[8] Mao, X. et al. Dual attention matching network (Dual-AMN). *WWW*, 2021.

[9] Sensoy, M. et al. Evidential deep learning for classification uncertainty. *NeurIPS*, 2018.

[10] Sun, J. et al. Think-on-Graph (ToG). *ICLR*, 2024.

[11] Sun, Z. et al. Bootstrapping entity alignment (BootEA). *IJCAI*, 2018.

[12] Wang, Z. et al. GCN-Align. *EMNLP*, 2018.

[13] He, G. et al. ChatKBQA. *ACL*, 2024.

[14] Ji, S. et al. A survey on knowledge graphs. *IEEE TNNLS*, 2022.